

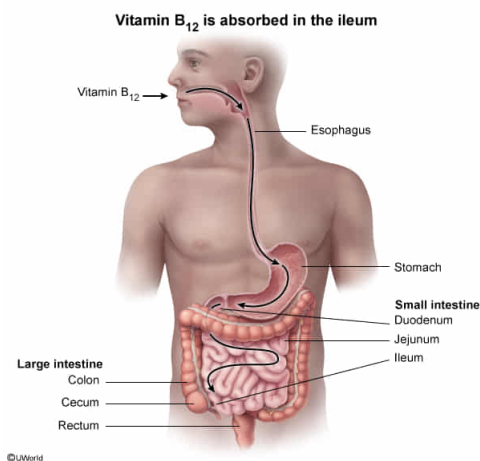
Vitamin B₁₂ is absorbed by the region of the digestive system that immediately precedes the cecum. Given this, vitamin B₁₂ absorption most likely takes place in the:

- ☐ A. duodenum. [12%]
✓ ☒ B. ileum. [52%]
☐ C. jejunum. [18%]
☐ D. colon. [16%]

Correct

52% answered correctly

Explanation:



Ingested food moves from the mouth through the muscular **gastrointestinal tract** due to the actions of **peristalsis**. The gastrointestinal tract consists of the following **components**:

1. **Esophagus** is the tube through which ingested food, becomes a compact food mass (ie, bolus), and moves from the mouth, through the cardiac (lower esophageal) sphincter, and into the stomach.
2. **Stomach** is the sac in which the food bolus is chemically (ie, via **stomach cell secretions**) and mechanically digested, creating a semiliquid substance (ie, chyme) that moves through the pyloric sphincter into the small intestine.
3. **Small intestine** is the long, narrow tube consisting of **three subdivisions** (duodenum, jejunum, ileum) in which **digestion of macromolecules** is completed and the resulting useful substances are absorbed. Digestion of lipids in the small intestine is aided by **bile**, which is a nonenzymatic solution produced by **liver** cells and stored in the gallbladder. In addition, the **pancreas** secretes digestive enzymes into the small intestine along with bicarbonate, which assists in **neutralizing** the acidic chyme arriving from the stomach. Material unable to be digested passes to the large intestine through the ileocecal sphincter.
4. **Large intestine** is the wide tube consisting of **three subdivisions** (cecum, colon, rectum) in which water and electrolytes are absorbed from chyme to form solid feces.

The question states that vitamin B₁₂ is absorbed in the region of the digestive system that immediately precedes the cecum. Immediately before chyme enters the cecum, it is found in the ileum (final segment) of the small intestine. Therefore, vitamin B₁₂ absorption most likely occurs in the **ileum**.

(Choice A) The duodenum is the *initial* segment of the small intestine and does *not* immediately precede the cecum of the large intestine.

(Choice C) The jejunum immediately precedes the ileum of the small intestine, *not* the cecum of the large intestine.

(Choice D) In the large intestine, the colon follows (*not* precedes) the cecum.

Educational objective:

The small intestine's segments include the duodenum, jejunum, and ileum. These segments of the digestive system are responsible for digestion of ingested food and absorption of useful nutrients.

Time spent:0 sec

Subject:
Biology

QID:403240

System:
Digestion and Excretion